occurred about 2.5 billion years ago, at the end of the Archean Eon, when oxygen first appeared on Earth. At this time, Earth was a hot, acidic place in which methane-producing bacteria were the only life that thrived. Bacteria capable of photosynthesis then evolved, which created an atmosphere that was conducive to more complex forms of life. Eventually, the equilibrium conditions that exist on Earth today were established.



In the Gaia hypothesis, Earth, the only known planet to support life, is itself a “superorganism,” where the sea, land, and atmosphere work together to maintain the right living conditions.